Multimodal Emotion Recognition in DUUI

Praktikumsplan

1 Issue of the Task

Emotion recognition in video streams has traditionally relied on isolated, unimodal analyses (e.g., analyzing only facial frames). Initial testing with classic Vision Transformers (e.g., ViT-Base models trained on FER-2013 and AffectNet) revealed strong contextual biases on speech-heavy material—head posture and frontal framing artificially skewed predictions, resulting in roughly 75% "sad" classifications on parliamentary footage. This motivates extending the analysis to multiple modalities, but is not the main research focus of this project.

Emotion annotations are to be produced from three modalities —face/action units, audio prosody, and text-based emotion recognition on the transcript (body pose is set aside, see Section 3). Suitable models for each modality are largely available off-the-shelf.

The main contribution lies in the **pipeline construction within DUUI**: how these heterogeneous components are composed into a coherent multimodal pipeline. The central design challenges are:

- **Time axis handling** across modalities — text-based (sentences as segment anchors) vs. frame/time-based (uniform windows, shots, or single frames).
- **Video-specific properties** — variable frame rates, resolutions, codec quirks, and the corresponding metadata bookkeeping required for reproducible analysis.
- **TypeSystem design** — a UIMA annotation schema tailored to our multimodal emotion use case, with shared segment anchors that enable cross-modal references.

2 Available Resources

Framework: DUUI / MU-DUUI with existing components: `duui-whisperX` (transcription with word-level timestamps), `duui-yt_dlp` (video download), spaCy containers (tokenisation, sentence splitting). UIMA TypeSystem with `MultimediaElement`, `AudioToken`, `VideoToken` as a basis. A Bundestag corpus reader component is either available or implemented as part of Phase 1.

Annotation schemes per modality:

- **Face:** categorical (Ekman 6/7/8), dimensional (Valence/Arousal), Action Units (FACS, ∼40 AUs).
- **Audio/Prosody:** categorical (Ekman variants or IEMOCAP set) or dimensional (V/A, optionally Dominance).
- **Text:** emotion classification (e.g. Ekman categories) or dimensional Valence/Arousal repre-

sentations.

Body pose is deliberately omitted from the initial scope (see Section 3).

Tech stack: Python, HuggingFace Transformers, Docker, REST. Concrete model choices in Phase 1.

VLMs (side track): Phi-4, Qwen-VL as comparison baselines.

Corpus: Bundestag plenary protocol videos as the main evaluation corpus; control corpora (talk shows, vlogs from YouTube).

3 Approach

The project starts from a **sentence-based time alignment** approach: sentences derived from the WhisperX transcript serve as the common anchor for all annotations. In parallel, we explore how the same component architecture can be built for the text-free case (video plus audio only). The emphasis throughout is on **variable, reusable components** that can be recombined into different pipelines without code changes.

Core pipeline (sentence-anchored):

`Reader` → `WhisperX` → `spaCy` → `SentenceAligner` → 3× `analysis` → `Writer`

The Reader is responsible for ingesting all modalities of one input (video, audio, text, metadata) and writing them into the appropriate UIMA views. Different corpora (Bundestag vs. YouTube control material) require different readers, but the downstream pipeline is identical. The `SentenceAligner` joins spaCy sentence spans with WhisperX `AudioToken` timestamps to produce `SentenceSegment` annotations carrying both text offsets (`begin`/`end`) and time intervals (`timeStart`/`timeEnd`). All analysis components consume the same segment pool.

An open design question, decided during Phase 1, is how the Reader, Writer, `SentenceAligner`, and `VideoSegmenter` are realised. Ideally each of them is implemented as a full DUUI Docker component, but external pre- or side-processing (producing or modifying JCas documents outside the pipeline) remains an option where it makes more sense.

Analysis components:

- **Face/V/A + FACS:** per segment, sample frames and classify Valence/Arousal and FACS Action Units.
- **Audio Emotion:** per segment, extract the audio clip and run a speech emotion model (categorical or V/A).
- **Text Emotion Recognition:** per segment, run a German or multilingual emotion classifier on the transcript text.

Pose and body language are deliberately omitted from the initial scope — interesting in principle, but they add complexity (separate model family, pose-to-emotion derivation is not well-established off-the-shelf) without being on the critical path of the alignment question.

Text-free variant. In parallel, we investigate how to handle pipelines without a transcript anchor — i.e. for video-only or video-plus-audio input where no sentence boundaries are available. Here, segments have to be constructed artificially in some frame- or time-based way; a dedicated component (e.g. a `VideoSegmenter`) could take care of this. The TypeSystem must accommodate such artificial segments alongside the sentence-based ones, and the analysis components should remain usable in either regime.

Side project — VLM comparison. How much do VLM predictions differ from the classical pipeline's output when prompted to produce comparable outputs (V/A, Ekman classes, or combinations thereof)? On selected segments, Phi-4 or Qwen-VL are queried with frame, audio clip, and transcript text together. Results are stored as ordinary `EmotionAnnotation` entries and compared to the classical components. Stretch goal, not on the critical path.

4 Research Questions

Beyond the per-segment annotations themselves, the project asks how the resulting data can be aggregated and interpreted at three temporal levels:

- **Micro:** emotion per individual segment (sentence or, in the text-free variant, frame/window). Cross-modal agreement and divergence at the local level.
- **Meso:** emotion per speaker section — duration of emotional states, volatility, dominant patterns within one speaker's contribution.
- **Macro:** emotion across a full session or day — shifts between speakers, escalation or contagion patterns over time.

The TypeSystem and aggregation logic must support all three levels, even if not all of them are explored equally deeply during the project.

5 Phases

- **Phase 1 — Building the sentence-anchored pipeline.** Build the components needed for the sentence-anchored pipeline end to end: Reader, `SentenceAligner`, and the three analysis containers (Face/V/A+FACS, Audio Emotion, Text Emotion). The TypeSystem grows alongside, since the required types emerge directly from what the components write. Goal: a runnable pipeline on a small Bundestag sample.
- **Phase 2 — Pipeline evaluation and redesign for variability.** Review the working pipeline: which parts are tied to the sentence-anchor assumption, which are already generic, and how to make them as variable as possible (generic segment supertype, components that work with both `SentenceSegment` and frame/time-based segments). Prototype a `VideoSegmenter` as a test case for the text-free variant.
- **Phase 3 — Data evaluation.** Cross-modal aggregation on a larger Bundestag sample at the micro/meso/macro levels, complemented by runs on the other mentioned sources (talk shows, YouTube vlogs) to see how the pipeline behaves outside the parliamentary setting.
- **Phase 4 (side project) — VLM comparison.** Query Phi-4 or Qwen-VL with combined modalities on a sample of segments and compare to the classical components' output.

6 Expected Deliverables

- A TypeSystem extension for multimodal emotion annotations in MU-DUUI, centred on the `SentenceSegment` anchor.

- Three analysis components, plus a Reader, a `SentenceAligner`, and possibly a `VideoSegmenter` — each either as a full DUUI Docker component or as lightweight glue code, depending on the Phase 1 design decisions.
- A runnable sentence-anchored pipeline and an evaluation report on cross-modal agreement on Bundestag data.
- (Optional) A comparison between sentence-based and frame-based anchoring strategies.

AI (Claude Opus 4.7) was used to format this text into a readable and well-structured LaTeX layout.

Danke für Ihre Beschreibung. Insgesamt ein guter erster Aufschlag. Nachstehend meine Kommentare im Generellen:

- Es fehlt an Quellen und Referenzen.
- Mir ist relativ viel AI in diesem Text, zumindest was die Beschreibung angeht; bitte nicht mehr verwenden.
  - Ein TypeSystem muss zu Beginn erstellt werden, denn das gehört zum Design.
- Wie wollen Sie mit mehreren Personen im Video / Frame umgehen?
- Wo ist die Verbindung zwischen den Komponenten, wie sieht das Datenmodell aus?
- Wie kommen Sie an die Daten heran, woher kommen Ihre Golddaten als Vergleich?
- Wie wollen Sie das ganze Evaluieren?
- Ein Zeitplan fehtl.

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